Aarya Aamish Bhatt

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Education

University of California, Berkeley – Haas School of Business **Jan 2027 - Jan 2028**
Master of Financial Engineering | GRE Quant: 170/170 *Berkeley, CA*

Birla Institute of Technology & Science, Pilani **Nov 2020 - Jun 2025**
Dual Degree: M.Sc. (Hons.), Economics and B.E. (Hons.), Computer Science *Goa, India*
Achievements: CGPA 9.04/10 (Distinction) (US scale: 3.96/4.0), Recipient of the Merit Scholarship (Top 2% in university); Ranked in the Top 0.3% of 1.5M+ candidates in IIT-JEE; Recipient of INSPIRE Scholarship (Top 1%, Class 12, Govt. of India)

Skills & Achievements

Programming: Python(Pandas, NumPy,Scikit-Learn, TensorFlow), C++, Java, KDB/Q, SQL, R, Stata.
Mathematics: Probability Theory, Stochastic Calculus, Time-Series Analysis, Applied Econometrics, Linear Algebra, Numerical Methods, Partial Differential Equations, Numerical Analysis.
Finance & Quant: Derivatives Pricing & Risk Management, Options Greeks, Volatility Modeling, Portfolio Construction, Financial Risk Analysis, Security Analysis & Valuation, Data Structures & Algorithms, Machine Learning.
Tools & Certifications: Bloomberg Terminal, Bloomberg Market Concepts (BMC) Certification.
Achievements: 6-Time Teaching Assistant across Derivatives & Risk Management, Business Analysis & Valuation, and Security Analysis & Portfolio Management (80+ students/subject); mentored 850+ students in Probability & Statistics.

Professional Experience

Morgan Stanley **Mumbai, India**
Quantitative Investment Strategies (Associate) *Jul 2024 – Present*

- Contributed $750M+ in AUM to the trading desk by implementing intraday delta-hedging, momentum, and vol-target strategies for a London-based hedge fund in collaboration with sales and financial engineers.
- Engineered a volatility and exposure calculator module in Java leveraging Black-Scholes pricing to implement delta-hedging, theta/gamma-neutral, volatility-parity, and volatility-target strategies; the module was adopted across 15+ live indices, becoming a core component of the desk's strategy infrastructure.
- Derived & implemented a skew-delta calculation framework in Java, making Greek computations more realistic by accounting for volatility skew and improving P&L attribution for structured products.
- Built automated client reports in KDB/Freemarker to deliver real-time analytics(Greek P&L, real-time volatility scenario assessments, and index statistics), enabling clients to make timely trade decisions from live risk metrics.
- Worked closely with traders to identify and resolve pre-market risk issues, including Greek mismatches and index rebalancing anomalies, ensuring the desk's risk and P&L attribution were correctly stated before market open.

Oracle Financial Services Software Ltd. **Mumbai, India**
Digital Banking Solutions (Intern) *May 2023 – Jul 2023*

- Automated transactional data capture & audit logging workflows using **Oracle SQL triggers & stored procedures**, improving data accuracy and traceability across millions of transactions while reducing storage by **50%+**
- Developed analytical data models and dashboards in **Python and Excel** to monitor user activity and transaction trends, enabling data-driven performance evaluation for digital banking operations

Research Experience

Derivatives' Impact on Firm Valuation, Risk, and Stability Metrics -- *Independent Research | BITS Pilani*

- Found that hedging-oriented banks exhibit significantly lower earnings volatility and stronger solvency metrics (Tobin's Q, Z-score, Idiosyncratic risk) relative to speculation-oriented peers, using panel data regression across 90+ global banks sourced from Bloomberg.
- Segmented banks by derivatives usage intensity using ML-driven K-means clustering (validated with Elbow & Silhouette scores), isolating hedging vs. speculation behaviour as the key differentiator in firm stability.

NIFTY 50 Implied Volatility Surface & Delta-Neutral Signal-- *Independent Project*

- Constructed Nifty 50 implied volatility surface via Black-Scholes inversion across 6 weekly expiries and 20+ strikes; fitted a smoothed surface using SVI parametrization to identify options trading rich or cheap relative to fair vol.
- Backtested a delta-neutral long-cheap/short-rich vol strategy (906 trades) with static delta-hedging, per-lot normalisation, and transaction costs; achieved OOS Sharpe of 4.37 and max drawdown of −1.78%, validated via walk-forward analysis on 2024 NSE data.

Interests: Quantitative Finance, Poker, Competitive Programming, Chess, Pickleball, Badminton, Formula 1.